

14-3d A Shift in Demand in the Short Run and Long Run

Now that we have a more complete understanding of how firms make supply decisions, we can better explain how markets respond to changes in demand. Because firms can enter and exit in the long run but not in the short run, the response of a market to a change in demand depends on the time horizon. To see this, let's trace the effects of a shift in demand over time.

Suppose the market for milk begins in a long-run equilibrium. Firms are earning zero profit, so price equals the minimum of average total cost. Panel (a) of [Figure 8](#) shows this situation. The long-run equilibrium is point A, the quantity sold in the market is Q_1 , and the price is P_1 .

Figure 8

An Increase in Demand in the Short Run and Long Run

Panel (a) shows a market in a long-run equilibrium at point A. In this equilibrium, each firm makes zero profit, and the price equals the minimum average total cost. Panel (b) shows what happens in the short run when demand rises from D_1 to D_2 . The equilibrium goes from point A to point B, price rises from P_1 to P_2 , and the quantity sold in the market rises from Q_1 to Q_2 . Because price now exceeds average total cost, each firm now makes a profit, which over time encourages new firms to enter the market. Panel (c) shows how this entry shifts the short-run supply curve to the right from S_1 to S_2 . In the new long-run equilibrium, point C, price has returned to P_1 but the quantity sold has increased to Q_3 . Profits are again zero, and price is back to the minimum of average total cost, but the market has more firms to satisfy the greater demand.


 Three pairs of graphs, each with a market graph and a firm graph, plot change in demand initially, in the short run, and in the long run. On the horizontal axis, the market graphs mark quantity (market) while the firm graphs mark quantity (firm). The vertical axis in all graphs mark price. In the two graphs titled "(a) Initial Condition," the market graph is as follows. Through point A, the intersection of the downward sloping demand curve, D_1 , and the upward sloping short-run supply curve, S_1 , the long-run supply curve is plotted as a horizontal line beginning at P_1 on the vertical axis and running parallel to the horizontal axis. A perpendicular dropped from point A touches the horizontal axis at Q_1 . In the firm graph, the upward sloping marginal cost curve intersects the lowest point on the average total cost curve. Through this point runs a line parallel to the horizontal axis and touching the vertical axis at P_1 marked in the market graph. The callouts in the graphs pointing at the intersections together read, "A market begins in long-run equilibrium ... with the firm earning zero profit." In the two graphs titled "(b) Short-Run Response," the market graph is the same as in "(a) Initial Condition" but with an additional downward sloping demand curve, D_2 , parallel to D_1 and cutting through S_1 at point B located above point A. A perpendicular dropped from point B touches the horizontal axis at Q_2 . A line passing through point B runs parallel to the horizontal axis, touching the vertical axis at P_2 . In the firm graph, through the intersection of the upward sloping marginal cost curve and the horizontal line from P_2 , as marked in the market graph, a perpendicular is dropped to touch the average total cost curve below at a point from where a line runs parallel to the horizontal axis to touch the vertical axis between P_1 and P_2 . The rectangular area between the horizontals is shaded. The callouts in the graphs, one pointing at point B and other at the shaded rectangle, together read, "But then an increase in demand raises the price ... leading to short-run profits." In the two graphs titled "(c) Long-Run Response," the market graph is the same as in "(b) Short Run Response" but with an additional upward sloping supply curve, S_2 , parallel to S_1 and cutting through the long-run supply curve at point C, located further away from the vertical axis than point A. Curve D_2 also passes through point C. A perpendicular dropped from point C touches the horizontal axis at Q_3 . The firm graph is the same as in "(a) Initial Condition." The callouts in the graphs, one pointing at S_2 and the other at the intersection of curves in the firm graph, together read, "When profits induce entry, supply increases and the price falls ... restoring long-run equilibrium."

Now suppose scientists discover that milk has miraculous health benefits. As a result, the quantity of milk demanded at every price increases, and the demand curve for milk shifts outward from D_1 to D_2 , as in panel (b). The short-run equilibrium moves from point A to point B; as a result, the quantity rises from Q_1 to Q_2 , and the price rises from P_1 to P_2 . All of

the existing firms respond to the higher price by increasing the amount they produce. Because each firm's supply curve reflects its marginal-cost curve, how much each firm increases production depends on the marginal-cost curve. In the new short-run equilibrium, the price of milk exceeds average total cost, so the firms are making positive profit.

Over time, this profit encourages new firms to enter. For example, some farmers supplying other products may switch to producing milk. As the number of milk suppliers grows, the quantity supplied at every price increases, the short-run supply curve shifts to the right from S_1 to S_2 , as in panel (c), and this shift causes the price of milk to fall. Eventually, the price is driven back down to the minimum of average total cost, profits are zero, and firms stop entering. Thus, the market reaches a new long-run equilibrium, point C. The price of milk has returned to P_1 , but the quantity produced has risen to Q_3 . Each firm is again producing at its efficient scale, but because more firms are in the dairy business, the quantity of milk produced and sold is higher.

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